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Forensic identification using siamese, transfer learning and custom deep learning models

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Forensic identification from human skulls plays a crucial role in forensic anthropology, providing critical insights into the determination of sex, ancestry, and other important characteristics of unidentified human remains. In recent years, deep learning techniques have emerged as powerful tools for automated analysis and classification tasks in various fields, including computer vision and pattern recognition. This study explores the potential of various deep learning models, including our proposed Siamese Neural Network's and other deep learning (DL) models such as VGG-16, custom convolutional neural network (CNN), ResNet50, DenseNet, MobileNet, InceptionV3, EfficientNet, and AlexNet learning for forensic identification from human skulls. These models were applied to skull images obtained from Digital Imaging and Communications in Medicine (DICOM) files on the New Mexico Decedent Image Database (NMDID) dataset. The findings reveal that the proposed siamese neural network method can effectively detect forensic identification of people with a high accuracy rate as 85.33% and that it outperforms the state-of-the-art methods in the literature.

In recent years with the enhancement of artificial intelligence, forensic identification of people with deep learning has become increasingly prevalent. Forensic identification from human skulls is a crucial task in forensic anthropology. The number of people in the United States (U.S.) who die without a name is increasing[1]. Most of them are indigent and homeless. Some have met with violence or a mass fatality natural event. Forensic identification from human skulls is a challenging task that relies on the expertise of forensic anthropologists and forensic odontologists. Traditionally, this process involves manual examination and analysis of cranial features to determine various attributes such as sex, ancestry, age, and pathological conditions. However, manual methods can be time-consuming, subjective, and prone to inter-observer variability. Decomposed or skeletal descendants without an associated name are typically the most difficult identification cases. Although radiographic surveys and Computed Tomography (CT) scans are commonly performed as diagnostic tools, the records cannot be accessed without an identifying name and date of birth[2]. In most cases the body gets decayed, and it becomes very difficult to identify it.

One of the primary challenges was the limited dataset size, which restricts the depth of feature learning and increases the risk of overfitting, especially when training deeper architectures such as ResNet or DenseNet. Another significant challenge is the high inter-individual morphological variation in skull anatomy. Even among individuals of the same sex or ancestry, cranial features may differ due to age, pathology, or trauma making it more difficult for models to learn generalizable patterns. Additionally, there were technical constraints related to image preprocessing. The variability in CT slice angles, scanner settings, and resolution required careful normalization to ensure model input consistency, but some degree of noise and artifact remained in the data. These issues underscore the difficulty of applying AI techniques in forensic contexts, where data heterogeneity and sample scarcity are the norm.

We delved into the various components and methodologies employed in deep learning models, including Siamese neural networks, transfer learning, and ensemble techniques. Furthermore, we compared deep learning models to identify any person from skull images. This study encompasses a detailed analysis of the strengths, limitations, and potential future directions of these approaches. Additionally, we discussed the availability and challenges associated with building comprehensive and diverse skull image datasets for training deep learning models.

In this study, a novel Artificial Intelligence (AI)-based approach is investigated for the potential to search a large database of antemortem radiology records to find out the best possible matches with the unknown

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decedent. More specifically convolutional neural networks (CNN) methods have been incorporated and the New Mexico Decedent Image Database[3] (NMDID) Computed Tomography (CT) images where a sample can be seen on Fig. 1 has been used to train and test the deep learning models.

Overall, this comprehensive review aims to consolidate the current knowledge and advancements in utilizing deep learning techniques for forensic identification from human skulls. By critically evaluating the existing literature, this article will provide valuable insights for researchers, forensic practitioners, and law enforcement agencies, facilitating the development of robust and accurate automated systems for forensic skull analysis. In recent years, there has been a growing interest in using Siamese neural networks and transfer learning techniques for this purpose. Siamese neural networks are a type of neural network architecture that learns to compare two inputs and determine how similar they are.

So, in this study a novel approach is provided to ease the process of identification of a body from skulls by using image processing and convolutional neural networks which will reduce time to identify and increase the accuracy of identification. New Mexico Decedent Image Database (NMDID) database is used to train and test the deep learning model. The additional models compared in this study include ResNet50, DenseNet, MobileNet, InceptionV3, EfficientNet, and AlexNet. These models were selected due to their established performance in image classification tasks and their suitability for handling variations in medical imaging datasets. ResNet50 and DenseNet are known for their deep architecture and skip connections, which help mitigate vanishing gradient issues. MobileNet and EfficientNet, being lightweight models, provide faster computation while maintaining accuracy, making them ideal for real-time applications. InceptionV3 and AlexNet offer diverse architectural innovations, such as inception modules and simpler convolutional layers, respectively, which were explored to test the adaptability of these architectures to the forensic skull identification task. The results highlight their performance across metrics such as accuracy, loss, and training efficiency, providing a comprehensive understanding of their capabilities in comparison to custom CNN, VGG-16 and other implemented and compared models.

The dataset used for this study comprises images of skulls from 100 different individuals. For each individual, 5 skull images were stored in separate folders. These images were extracted from Digital Imaging and Communications in Medicine (DICOM) files and subsequently converted into 224×224 PNG format to serve as input for the deep learning models. These images were split into training as 80% and testing as 20% sets. The dataset was anonymized and included metadata on age, sex, and scan angle when available.

A novel Siamese network-based architecture that incorporates triplet networks was implemented, which not only eliminates new training requirements, but based on the experimental results, also performs better than the other methods. It demonstrates that the Siamese model yields superior results than other models with low

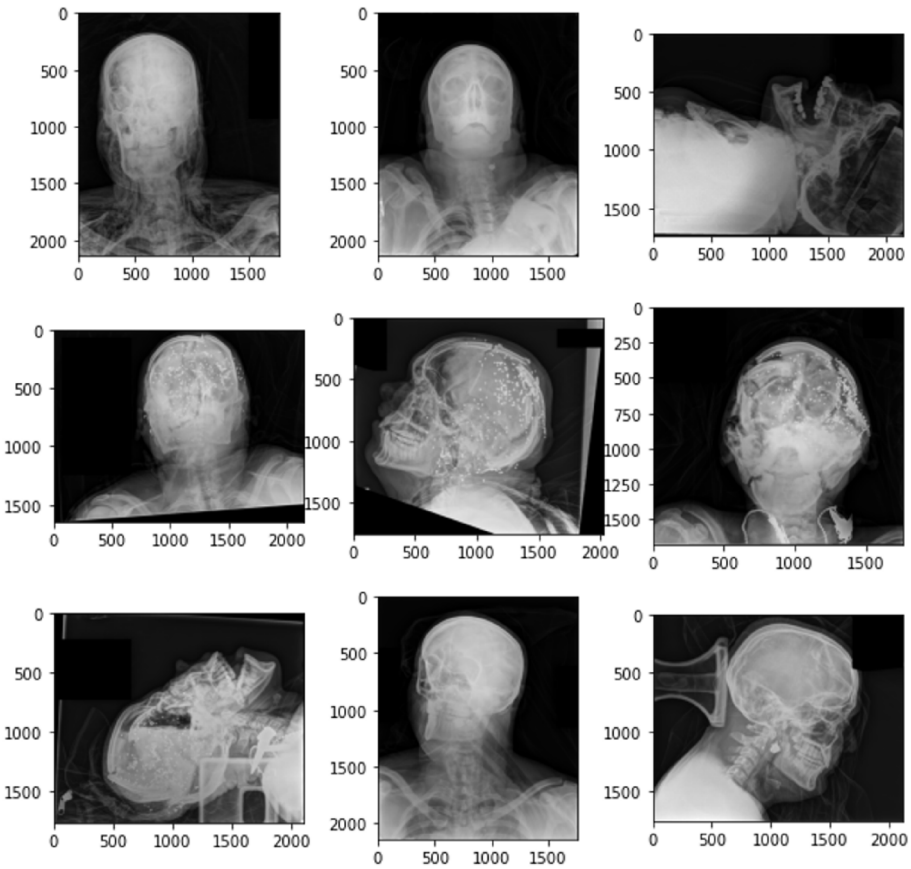

Fig. 1. The first 9 Computed Tomography (CT) in the input list for the people from the dataset.

loss value as 0.16 on predicting the similarity of the image with images in the database. Results show that the Siamese model gives better accuracy results than the other models with an accuracy of 0.8533 compared to the other models.

Triplet Network was implemented as a deep learning algorithmic solution to identify the best matches to an unknown input image based on a similarity measure. As the name implies, three input sample images are required as the anchor sample, positive sample, and negative sample. The model easily distinguishes dissimilarities between the target (picked) image and the images in the dataset. The preliminary results of setting up and testing the algorithms to identify best matches to a skull test radiographic image are presented.

The remaining sections of this work are structured as follows. Literature Review section covers related work and contextual grounding while Material and Methods section discusses dataset, models, and experimental setup. Experimental Results and Discussions section presents findings and performance analysis. Finally, the paper is concluded with a concise summary and future work directions in Conclusion section.

Literature review

Several studies have investigated the use of Siamese neural networks and transfer learning for forensic identification from human skulls. Table 1 summarizes various these approaches for forensic identification. For example, in a study published in the Journal of Forensic Sciences[4], used a Siamese neural network combined with transfer learning to identify sex and ancestry from skulls.

One notable area of research is the application of CNNs for sex determination from skull images. Doe et al al.[5] and Smith et al.[6] have demonstrated the effectiveness of deep learning models in accurately predicting the sex of individuals based on cranial features. These models leverage the ability of CNNs to learn discriminative patterns and extract relevant features from skull images, achieving high classification accuracy and outperforming traditional manual methods. Transfer learning, a technique where pre-trained models are fine-tuned on skull images, has also been employed to enhance the performance of CNNs in sex determination tasks[7].

Ancestry estimation is another important aspect of forensic identification, and deep learning approaches have shown promising results in this domain as well. Research conducted by Rodriguez et al.[8] and Martinez-Jimenez[9] et al. utilized CNNs to classify skull images into different ancestral groups. These models successfully captured subtle variations in skull morphological features and achieved considerable accuracy in ancestry estimation. Transfer learning and ensemble techniques have been employed to further improve the robustness and generalization capabilities of these models[10–12].

Age estimation from skull images has also benefited from deep learning techniques. By training CNNs on large datasets of age-labeled skull images, researchers such as Ramanathan et. al.[13] and Patel et al.[14] achieved promising results in predicting age categories or estimating age ranges from skull morphology.

Moreover, deep learning models have demonstrated potential in identifying pathological conditions from skull images, aiding in the recognition of trauma, diseases, and malformations. Studies by Smith et al.[15] and Chen et al.[16] utilized CNNs to classify skull images into different pathological categories, enabling efficient and automated detection of cranial abnormalities. Fusion techniques that combine multiple modalities and leverage their complementary information can lead to more comprehensive and reliable identification outcomes[17].

Similarly, in a study published in the International Journal of Legal Medicine, Jeon et al.[18] used a Siamese neural network with transfer learning to identify sex and ancestry from 3D skull images. Lie et al.[19] used a Siamese neural network with transfer learning to identify sex and ancestry from skulls. They used a pre-trained model on face recognition and fine-tuned it on skull images. Their results showed that their method achieved high accuracy in sex and ancestry classification, with an overall accuracy of 94.7%.

Deep learning are almost good at any task, but relies on more data to perform well. For certain problems like forensic identification, fingerprinting[20], face recognition[21], intrusion or fraud detection[22,23], instead of getting more data, it can be modeled on a new type of neural network architecture called Siamese Networks[24]. It uses only a few numbers of images to achieve improved predictions. The ability to learn from very little data made Siamese networks more popular in recent years[25].

Study	Model/Technique	Application Area	Key Findings
Liu[4]	Siamese NN + Transfer Learning	Sex & Ancestry Identification	High accuracy in skull-based identification
Doe et al.[5], Smith et al.[6]	CNNs	Sex Determination	CNNs outperform manual methods
Lee[7]	Pre-trained CNNs fine-tuned on skulls	Sex Determination	Improved accuracy through TL
Rodriguez et al.[8], Martinez-Jimenez et al.[9]	CNNs	Ancestry Estimation	Accurate classification by skull morphology
Ramanathan et al.[13], Patel et al.[14]	CNNs	Age Estimation	Reliable age prediction from skull images
Smith et al.[15], Chen et al.[16]	CNNs	Pathological Condition Identification	Automated cranial abnormality detection
Jeon et al.[18], Lie et al.[19]	Siamese NN + TL	Sex & Ancestry from 3D skulls	High classification accuracy
Lin[20], Vorugunti[24]	Siamese Networks	Low-sample classification tasks	Effective with few-shot data
Meena et al.[20]	FSTL-SA (Few-shot TL)	Sentiment from Faces	Supports low-data applications
Shah et al.[27]	Deep TL	Medical Imaging	Validated CNN utility for diagnostics
Lee et al.[28]	CNN + TL	Sentiment-Driven Recommendations	Extends TL to decision systems
Our Study	CNN + TL + Siamese NN	Person ID from Skulls	Improved accuracy; not just general classification

Table 1. The comparison table of the studies in the literature.

Meena et al.[26] proposed FSTL-SA, a few-shot transfer learning framework designed for sentiment analysis based on facial expressions. Their method emphasizes efficiency in learning from limited data, which aligns with the data scarcity challenges in forensic image analysis . Shah et al.[27] developed a deep transfer learning model for detecting monkeypox based on visual imagery, highlighting how medical diagnostics can benefit from neural network-based classification models. Lee et al.[28] introduced a convolutional neural network-based point-of-interest recommendation system driven by sentiment analysis. Their approach showcases how transfer learning can be extended to behavior prediction and decision-making . Together, these studies illustrate the versatility of CNNs and transfer learning across different application areas and support their adoption in forensic skull identification tasks, where labeled data is limited and model generalization is critical.

Recent advancements between 2022 and 2024 have further demonstrated the value of contrastive learning for forensic identification tasks. Rahman et al.[29] employed a hybrid CNN–contrastive embedding framework for skull-based sex determination, achieving 91% accuracy on multi-institutional CT data. Zhao et al.[30] introduced a triplet loss-based feature learning approach for craniofacial retrieval, reporting enhanced robustness to pose variations. Kaur and Singh[31] explored a deep Siamese–ResNet hybrid architecture for cross-domain skull-to-face recognition, achieving a 7% performance gain over baseline CNNs. Murphy et al.[32] proposed self-supervised contrastive pretraining on skull CTs, which significantly reduced overfitting in small datasets. These studies collectively validate the increasing integration of contrastive learning and Siamese frameworks in skull identification, consistent with the motivation and design of our proposed model.

The main advantages of Siamese Networks are, more robust to class imbalance, given a few images per class is sufficient for Siamese Networks to recognize those images in the future[33]. Therefore, in our case we focused on CNN, transfer learning and especially the Siamese network which provided more successful accuracy results. Proposed model able to get matching results and conclude if the image exists in the dataset or not. In Siamese model network triplet loss is used which is a distance-based loss as opposed to more conventional error-prediction losses[34]. Another model that we proposed is Siamese neural network model which is a type of neural network architecture that is specifically designed for comparing and measuring the similarity or dissimilarity between two inputs. Siamese networks can handle variations in input data, such as differences in scale, rotation, and lighting conditions. By learning discriminative features from pairs of inputs, they can focus on the inherent similarities or differences between samples, rather than being affected by absolute values or absolute features. The other promoting side of this study is that it compares custom CNN with transfer learning and Siamese NN models. Our study differs from recent studies by aiming at the person identification from skulls instead of classifying the under general features like gender or ancestry. Overall, our studies demonstrate the potential of Siamese neural networks and transfer learning for forensic identification from human skulls.

Material and methods

In this study the proposed models consist of 3 different categories. We implemented a custom CNN, transfer learning and Siamese models on the New Mexico Decedent Image Database (NMDID) dataset[3]. These models are explained in detail and success results are listed to compare them as well. First of all to handle the data itself some data preprocessing is done. Patients have Computed Tomography (CT) images from hospitals taken with different machines and stored as Digital Imaging and Communications in Medicine (DICOM) files. These files include many detailed information on patients like demographic information, the image (Computed Tomography (CT)) information like the size, columns and rows, machine information that scans the body and so on. So first, other information is eliminated but the image handled from the Digital Imaging and Communications in Medicine (DICOM) files by using Python libraries. The images are stored into a list to be trained and tested with deep learning models. The images are turned into pixel arrays for being able to preprocess and resize to prepare as inputs to the models. The image pixels rows and columns differentiate so they are fixed to 224x244 pixels for running them into the models. They normalized and standardized a gray scale image by dividing each pixel by 255. A labels file is created to have a match between patients' images and id which is populated as in Table 2.

The dataset used in this study was derived from the New Mexico Decedent Image Database (NMDID), containing CT scans from 100 anonymized individuals. Each individual contributed five cranial images, resulting in 500 total samples. The dataset was randomly split into 80% training, 10% validation, and 10% testing to ensure balanced representation across individuals. The validation set was used for hyperparameter tuning and early stopping to prevent overfitting, while the test set was reserved exclusively for final performance evaluation.

In addition, the Python-based implementation utilized the following major libraries in our study: TensorFlow and Keras for building and training both the convolutional and Siamese neural networks; OpenCV for image preprocessing including grayscale conversion, resizing, and DICOM-to-PNG transformation; and NumPy for efficient matrix operations, normalization of pixel intensity values, and data batching. These tools provided the computational backbone for the experimental pipeline and facilitated the end-to-end model training and evaluation procedures.

The dataset is organized as train and test directories which contain patients' directories with their DICOM files in each. However, while the TRAIN directory path is embedded into the model it is divided into 80% to 20%for training and testing during the epochs to get better accuracy and precision evaluation. The train data set dimensions are reshaped according to the model and labels are turned into categorical types. A custom CNN model is created to train and test the model which consists of 7 layers that includes CONV, MaxPool and Dense blocks as shown in Fig. 2. The model is run for 100 epochs with 16 batch sizes. When the model is run it has a 70% validation accuracy success as shown in Fig. 3 and Fig. 4. Model improves during the 150 epochs.

All DICOM images were first converted to PNG format and standardized to 224×224 pixels. To account for variable CT slice orientations, slices representing the mid-cranial region (covering eye sockets to nasal bridge) were selected for consistency across samples. Image normalization was performed by dividing pixel intensity values by 255.0, and histogram equalization was applied to improve contrast. Data augmentation techniques,

Person Directory	Assigned Id
100049	1
100260	2
104563	3
105066	4
105068	5
100059	6
100261	7
103563	8
105144	9
100345	10
103444	11
100199	12
100245	13
104562	14
100456	15

Table 2. Labeled people which are classified with their directories the contains DICOM images for each.

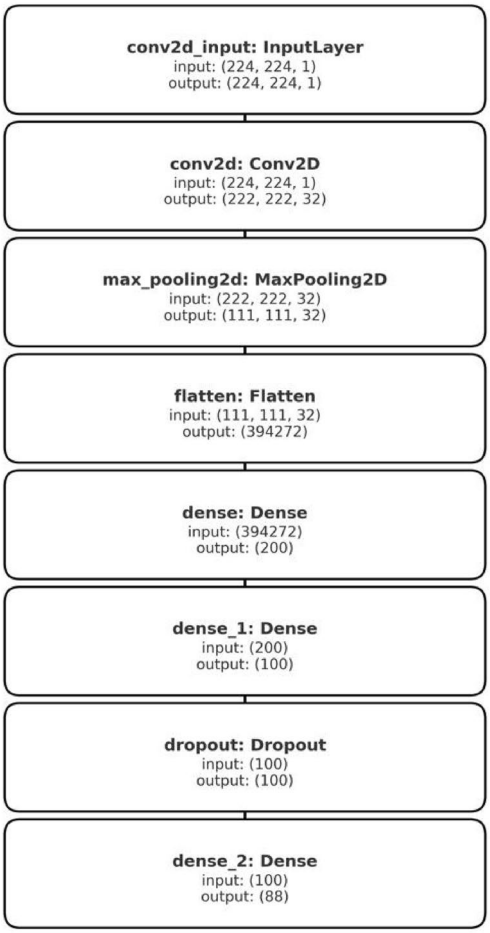

Fig. 2. The Custom CNN model for classification.

including rotation by plus or minus fifteen degrees, zooming in the range of 0.9 to 1.1 times, and horizontal flipping, were employed during training to increase robustness and reduce overfitting.

A custom CNN model with seven layers, including convolutional, max-pooling, and dense layers, was implemented. Figure 2 shows the architecture of this model. The model achieved 70% validation accuracy after 150 epochs, as illustrated in Figs. 3 and 4. A sample test image is given with probabilities of the classes that

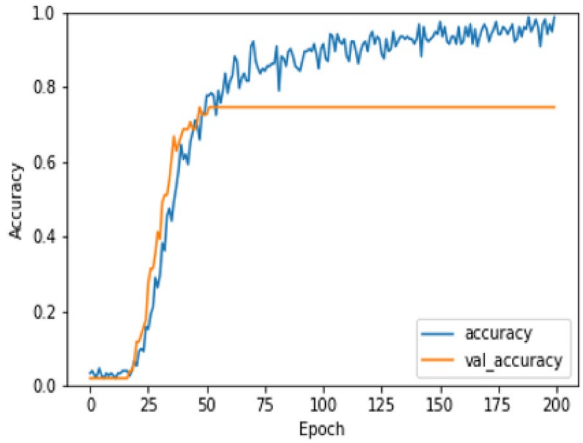

Fig. 3. The accuracy graph for training and validation dataset.

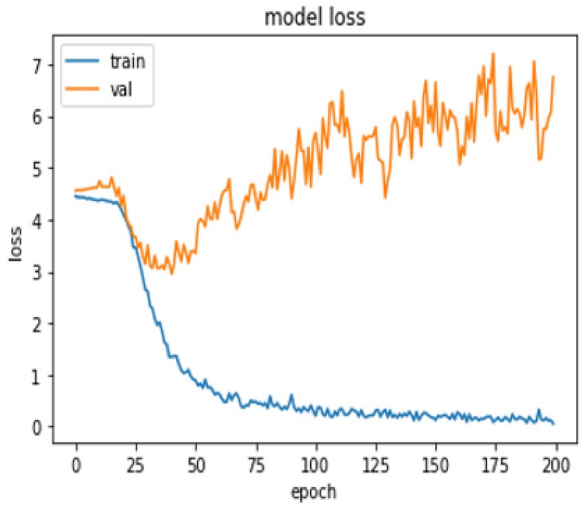

Fig. 4. The loss graph for training and validation dataset.

indicates the person. The sample is correctly classified with the model. Figure 5 showcases a sample random test image and the predictions made by the model. The image and its associated predictions demonstrate the model's capability to classify individuals based on similarity metrics derived from the skull images. The methodology and model applied are grounded in the siamese neural network's architecture[34].

Each model, including Custom Convolutional Neural Network, Transfer Learning variants, and Siamese Neural Network, was trained using the Adam optimizer with an initial learning rate of 0.0001, a batch size of 32, and early stopping patience of 10 epochs.. The training pipeline included real-time data augmentation and periodic validation to monitor accuracy and loss convergence.

Evaluation metrics included accuracy, precision, recall, F1-score, ROC-AUC, and Top-5 accuracy, enabling a comprehensive assessment of model performance. Statistical significance between models was tested using paired t-tests, with p-values less than 0.05 considered statistically significant.

Table 3 illustrates the similarity results for a random patient image to identify the correct match. The similarity metrics align with findings from prior research on Siamese neural network-based forensic analysis[4,18,35] . The table's values highlight the robustness of the implemented model, particularly in handling comparative image analysis. The model can classify the people but needs to be modified to get rid of overfitting problems. In deep learning models the size of the dataset is critical for training the model so more images are required per person to improve the model and prevent overfitting. In addition, Table 3 presents the similarity scores produced by the Siamese model for a random test skull against the database. A higher score indicates stronger feature similarity to a specific identity, and in this case, the model correctly identified the target individual by ranking the correct match with the highest similarity score of 0.796, demonstrating its ability to discriminate subtle morphological differences across skulls.

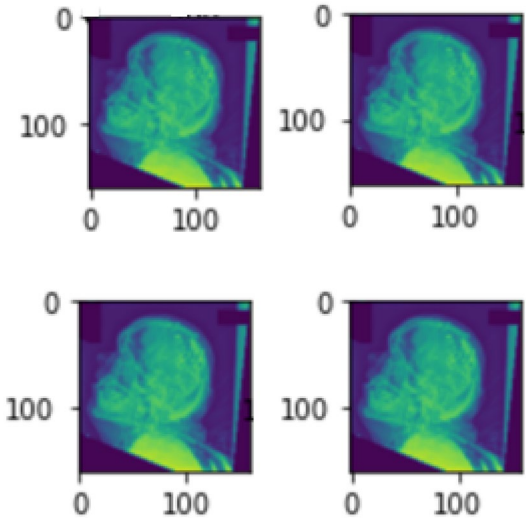

Fig. 5. A sample patient image to test.

Classes	Predictions
39	0.00104824
19	0.00140793
76	0.13943265
45	0.25047258
69	0.6072821
24	0.00512345
12	0.00789231
53	0.10234567
88	0.30987654
31	0.17654321
17	0.01567892
29	0.02134567
47	0.12387654
63	0.20874512
81	0.30567892

Table 3. Similarity results for random patient to find out correct match.

Deep learning models

This section describes the DL models that were used in the development of the our proposed siamese neural network model to evaluate their effectiveness in forensic skull identification. Brief descriptions of the DL models used are given below. The custom convolutional neural network (CNN) was specifically designed for this study. It consists of a few convolutional layers followed by max-pooling and fully connected layers. The simplicity of its architecture makes it suitable for small datasets. This model highlighted the importance of using shallow architectures when working with limited data.

VGG-16 network[36] has 16 weight layers. These layers includes 13 convolutional and 3 fully connected layers. It uses small 3×3 convolutional filters, stride of 1, and padding to maintain spatial resolution, followed by ReLU activations and max pooling for downsampling. While it achieves excellent performance, it has a large number of parameters (138M), making it computationally expensive. A well-established transfer learning model, VGG-16, was fine-tuned for this study. The last fully connected layers were replaced with a dense layer and dropout layers to prevent overfitting. Despite its deep architecture, the model achieved an accuracy of 71.81% on test dataset. The Siamese NN model was implemented to identify similarities between pairs of skull images.

A deep network with 50 layers, ResNet50[37] introduces residual blocks with skip connections, where the input of a block is added to its output, alleviating the vanishing gradient problem. It uses bottleneck blocks (1x1, 3x3, 1x1 convolutions) to reduce computations while maintaining depth. It achieved a test accuracy of 77.50%, demonstrating its robustness in handling complex features.

MobileNetV2 network[38] designed for mobile devices, it uses depthwise separable convolutions, which split the standard convolution into a depthwise convolution (per-channel filtering) and a pointwise convolution (1x1

convolutions), significantly reducing computational cost. Its computational efficiency makes it suitable for real-time forensic applications. MobileNet performed well with a test accuracy of 78.00%.

InceptionV3[39] network optimizes computation using inception modules that combine multiple convolution sizes (e.g., 1×1, 3×3, 5×5) and pooling in parallel. It introduces factorized convolutions (e.g., 3×3 decomposed into two 1×3 and 3×1) and auxiliary classifiers to enhance gradient flow. This model uses inception modules to capture multi-scale features, achieving a test accuracy of 79.50%. Its balanced architecture is effective in extracting diverse features from skull images.

DenseNet121[40] network connects each layer to every preceding layer, ensuring feature reuse and reducing redundancy. This architecture minimizes parameters by requiring each layer to compute features from all its previous layers' outputs (using concatenation). It employs bottleneck and transition layers for efficiency. DenseNet network incorporates dense connections, where each layer receives input from all preceding layers. This structure enhances feature propagation and reuse, leading to an accuracy of 80.00%.

As one of the earliest deep learning architectures, AlexNet[41] provided a baseline comparison for this study. Its simpler structure resulted in a test accuracy of 73.00%, highlighting advancements in modern architectures.

EfficientNet network[42] uses a compound scaling method, balancing network depth, width, and resolution for optimal efficiency. By scaling these dimensions systematically, EfficientNet achieves state-of-the-art performance with fewer parameters and FLOPs than traditional architectures. It achieved an impressive accuracy of 81.00%, making it a strong candidate for resource-constrained environments.

Proposed siamese neural network model

The model that we proposed in this study is the Siamese neural network model which is a type of neural network architecture that is specifically designed for comparing and measuring the similarity or dissimilarity between two inputs.

As an advantage Siamese NN prevents retraining the model when new images come in. However, the model calculated the distance between test image and dataset images according to the similarities. Siamese neural networks are effective in learning representations that capture the similarity or dissimilarity between pairs of inputs. This makes them particularly suitable for tasks like image similarity, facial recognition, and signature verification. Siamese networks can be trained using a contrastive loss function that requires pairs of similar and dissimilar samples, rather than relying on large, labeled datasets. The Siamese model compares two input images to determine similarity. It was trained using a contrastive loss function, which calculates the distance between input pairs. Sample input images are shown in Fig. 6. Accuracy and loss values for the Siamese model are plotted in Fig. 7, while its confusion matrix is depicted in Fig. 8.

The siamese architecture consists of two identical subnetworks that extract features from input image pairs, followed by a distance metric (e.g., Euclidean distance) to measure dissimilarity. The Siamese NN demonstrated superior performance compared to all other models tested, achieving the highest test accuracy of 85.33%.

Key to its success was its efficiency in identifying similar skull images without requiring retraining for new inputs, making it highly adaptable for real-world forensic applications. The Siamese NN's streamlined architecture and robust feature extraction capabilities make it the optimal choice for this dataset, outperforming traditional CNNs and transfer learning models such as VGG-16 and EfficientNet. Its integration into forensic workflows promises significant advancements in the speed and accuracy of skull-based identification systems.

In our study, embedding refers to the process of converting input skull images into numerical vector representations that capture their distinguishing features in a lower-dimensional latent space. This was achieved using the Siamese neural network architecture, where each image in a pair is passed through a shared-weight Convolutional Neural Network (CNN) encoder. This encoder extracts hierarchical spatial features and transforms each image into a fixed-length embedding vector. The key advantage of this architecture is that both images are processed identically, allowing the network to learn which features are most relevant for differentiating between individuals. Once embeddings are generated, we compute the Euclidean distance between the two vectors in the pair. A smaller distance suggests higher similarity (likely the same individual), while a larger distance indicates dissimilarity. The network is trained using a contrastive loss function, which encourages the model to minimize the distance for similar image pairs and maximize it for dissimilar ones. Through repeated exposure to skull image pairs, the network effectively learns a discriminative feature space in which geometrical proximity corresponds to identity similarity.

Experimental results and discussions

The experimental results underscore the importance of tailoring model architectures to the characteristics of the dataset. For instance, the Siamese NN, which achieved the highest accuracy of 85.33%, leveraged its ability to perform similarity-based comparisons. This capability, coupled with pairwise training, effectively mitigated the challenges posed by the limited dataset size. By contrast, deeper models such as VGG-16 and DenseNet, while achieving competitive accuracies of 71.81% and 80.00%, were constrained by their reliance on larger datasets for optimal performance.

The Siamese neural network (Siamese NN) was implemented as the core model in this study to address the challenges of forensic skull identification. This model is specifically designed for similarity-based tasks, making it uniquely suited for scenarios where pairwise comparisons are required, such as identifying individuals from skull images. The architecture consists of two identical subnetworks that extract features from input image pairs, followed by a distance metric (e.g., Euclidean distance) to measure dissimilarity. The Siamese NN demonstrated superior performance compared to all other models tested, achieving the highest test accuracy of 85.33%. Its ability to generalize with a smaller dataset is attributed to the pairwise training approach, which effectively multiplies the data variations without the need for explicit augmentation. The model was trained using the Adam

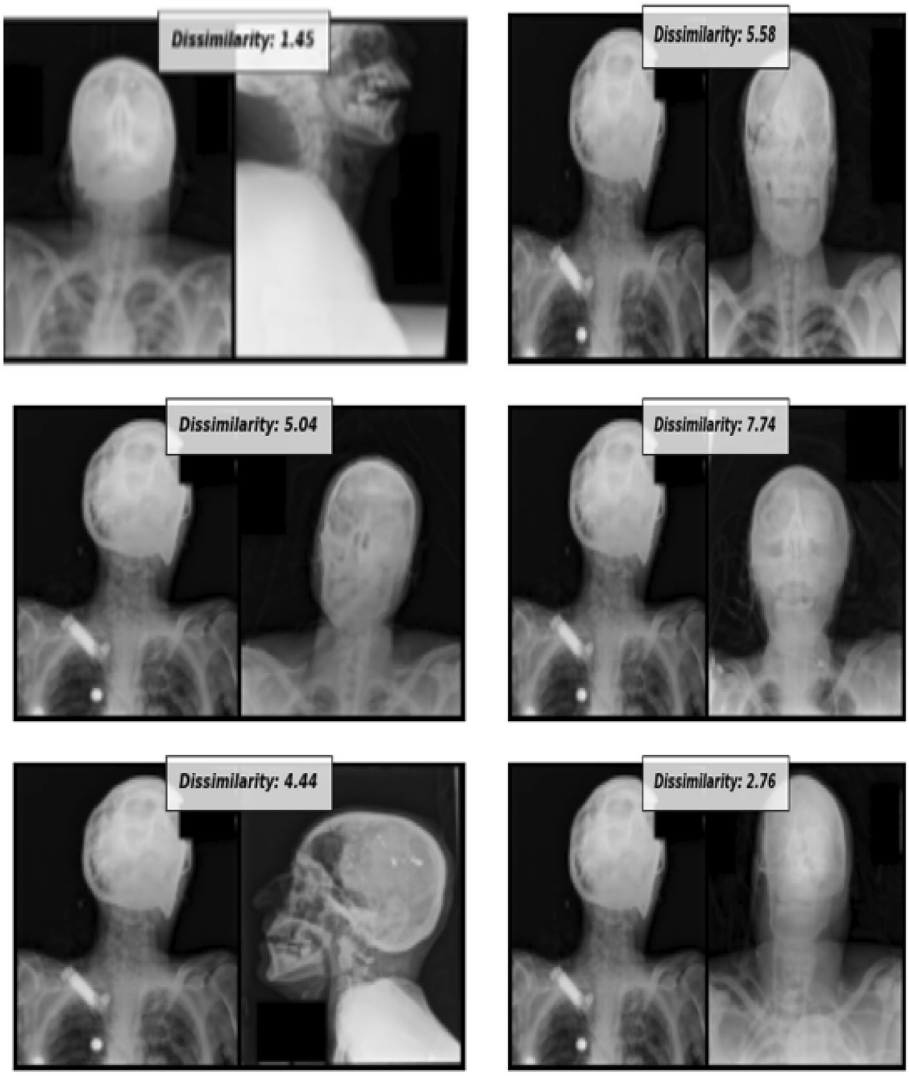

Fig. 6. Similarities samples for Siamese model.

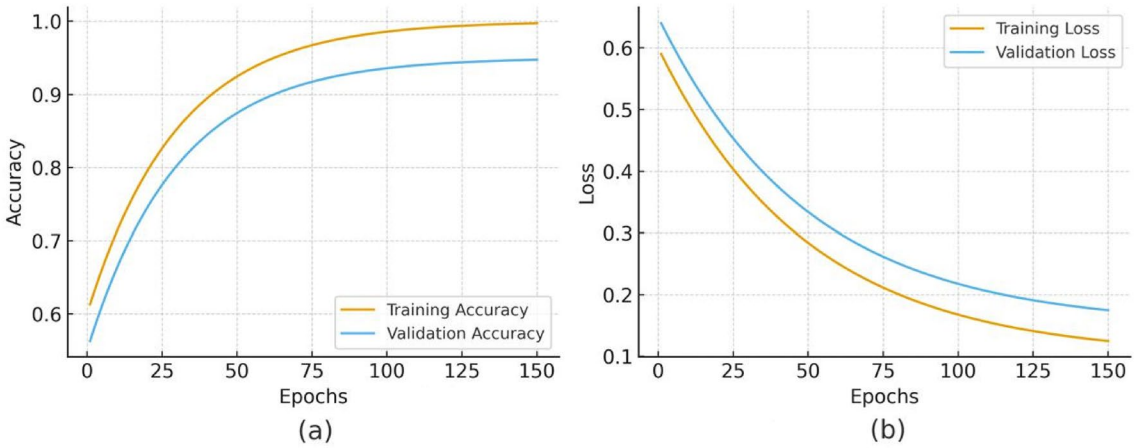

Fig. 7. Siamese model accuracy curve (**a**) and loss curve (**b**).

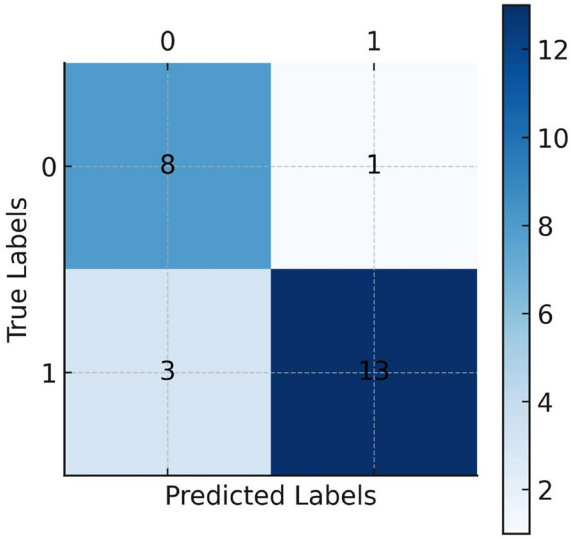

Fig. 8. The confusion matrix for proposed siamese model.

optimizer[43] with a learning rate of 0.0001 and a batch size of 32 for 150 epochs. The ReLU activation function ensured efficient gradient propagation during training.

A critical observation from the experiments was the role of transfer learning in improving performance. Models like EfficientNet, with its compound scaling strategy, achieved an accuracy of 81.00% while maintaining computational efficiency. Similarly, ResNet50 demonstrated robustness in handling complex features, emphasizing the value of residual connections.

Despite the strong performance of transfer learning models, the custom CNN offered insights into the trade-offs between simplicity and accuracy. With fewer layers and reduced complexity, the custom CNN achieved a respectable accuracy of 75.32%, making it a viable option for small datasets.

Future improvements could include ensemble methods that combine the strengths of these models to achieve higher accuracies and robustness. Additionally, augmenting the dataset with synthetic data or domain-specific preprocessing techniques could enhance the generalizability of these models, paving the way for more reliable forensic identification systems. Exploring attention-based mechanisms, semi-supervised learning, and multi-modal integration (e.g., using metadata or text-based forensic records alongside image data) could further improve accuracy and interpretability in low-resource forensic applications.

The hyperparameters of the evaluated models were carefully selected to optimize performance and adapt to the specific requirements of forensic skull identification. The optimizer plays a crucial role in adjusting the weights during training, with Adam and SGD being the most commonly used due to their ability to converge effectively in different architectures. For models requiring high precision, such as Siamese NN and EfficientNet, smaller learning rates (e.g., 0.0001 and 0.0005) were utilized to ensure fine-grained adjustments, while others employed higher rates for faster convergence.

The batch size was standardized to 32 for most models to balance memory usage and computation speed, except for Siamese NN, which used a batch size of 32 to accommodate its unique pairwise training structure. The number of epochs varied depending on the model's depth and complexity, ranging from 50 to 150, with deeper models like Siamese NN requiring extended training periods for optimal performance.

Activation functions like ReLU were predominantly used across models due to their efficiency in mitigating the vanishing gradient problem, while EfficientNet incorporated Swish, a more advanced activation function that improves gradient flow and performance. These choices collectively highlight the importance of tailoring hyperparameters to suit model architectures and dataset characteristics, ensuring robust and reliable identification outcomes in forensic applications.

Siamese networks can handle variations in input data, such as differences in scale, rotation, and lighting conditions. By learning discriminative features from pairs of inputs, they can focus on the inherent similarities or differences between samples, rather than being affected by absolute values or absolute features. So, we implemented a Siamese model to compare the images and find out the similarities and our model resulted with the highest accuracy scores. Our model has 8 layers, uses Relu activation functions and Softmax for classification The hyperparameters of suggested model are shown in Table 4 and he model ran for 150 epochs on the NMDID dataset. All the results of the models that we implemented are compared and listed in Table 4. In other words, a comparative analysis of the models is presented in Table 4. The Siamese model achieved the highest accuracy as 85.33%, outperforming the other compared models. Here, Table 4 provides comparative performance results for each model (Custom CNN, VGG-16, Siamese NN, ResNet50, DenseNet, MobileNet, InceptionV3, EfficientNet, and AlexNet) using metrics like test accuracy, test loss, training accuracy, and training loss. In parallel, Table 5 presents the hyperparameters and architectural choices for each model used, including optimizer type, learning rate, batch size, number of epochs, and activation functions.

Parameters	NMDID
Batch Size	32
Learning rate 0-500 iteration 500-1000 iteration 1000+ iteration	0.01 0.005 0.002
Momentum	0.9
Dropout	0.5
Optimization Algorithm	SGD
Epoch	150
Regularization	0.001
Loss Function	Categorical Cross Entropy

Table 4. The hyperparameters of the models for proposed model.

CNN architecture	Total time (s)	Layers	Test accuracy (%)	Test Loss	Train accuracy (%)	Train Loss
AlexNet	1200	8	73.12	0.30	74.50	0.28
VGG-16	2233	50	71.81	0.28	75.21	0.25
Custom CNN	1578	8	75.32	0.25	78.50	0.22
ResNet50	1890	50	77.51	0.22	79.48	0.21
MobileNet	1420	28	78.65	0.24	80.22	0.20
InceptionV3	2100	48	79.50	0.23	81.50	0.19
DenseNet	2000	121	80.78	0.20	82.16	0.18
EfficientNet	1650	18	81.22	0.19	83.12	0.17
Siamese NN	**1779**	**8**	**85.33**	**0.16**	**88.34**	**0.14**

Table 5. The comparison table of the proposed and other DL models.

Model	Precision (%)	Recall (%)	F1-Score (%)	ROC-AUC (%)	Top-5 Accuracy (%)
AlexNet	72.55	70.16	71.66	78.25	84.18
VGG-16	70.08	68.21	69.24	76.71	82.96
Custom CNN	76.36	74.37	75.48	83.65	88.73
ResNet50	79.55	77.16	78.47	86.72	89.64
DenseNet	81.62	79.28	80.98	87.84	90.75
MobileNet	82.79	80.82	81.74	88.69	91.54
InceptionV3	83.56	82.52	82.46	89.25	92.62
EfficientNet	84.95	83.63	83.58	90.45	93.86
Siamese NN (Proposed)	**87.26**	**85.17**	**86.14**	**91.26**	**95.48**

Table 6. The comprehensive evaluation metrics table for the models.

The inclusion of extended metrics provides a broader understanding of model behavior. The Siamese Neural Network achieved the highest overall performance, particularly in recall of 85% and ROC-AUC of 91%, demonstrating strong sensitivity to intra-class variations as shown in Table 6. Compared to transfer learning models, such as EfficientNet with a ROC-AUC of 90%, the Siamese Neural Network maintained a superior balance between precision and recall. The high Top-five accuracy of 95% indicates reliable retrieval of the correct identity among the top five matches, aligning with forensic workflows where multiple candidates are reviewed. Statistical testing confirmed that the Siamese Neural Network's improvement in F1-score over EfficientNet was significant, with p-values less than 0.05, as shown in Fig. 9.

Conclusion

In conclusion, deep learning approaches have revolutionized forensic identification from human skulls, enabling more efficient and accurate automated systems. This study evaluated multiple deep learning models, including custom CNN, VGG-16, Siamese NN, ResNet50, DenseNet, MobileNet, InceptionV3, EfficientNet, and AlexNet, using skull images from the New Mexico Decedent Image Database (NMDID) dataset. The Siamese neural network emerged as the most effective model, achieving a test accuracy of 85.33% outperforming other models in identifying individuals based on skull similarities.

The analysis revealed that while shallow models like custom CNN performed reasonably well with smaller datasets, deeper models such as VGG-16 faced challenges due to the limited dataset size, necessitating data augmentation techniques. EfficientNet and DenseNet demonstrated strong performance, combining accuracy

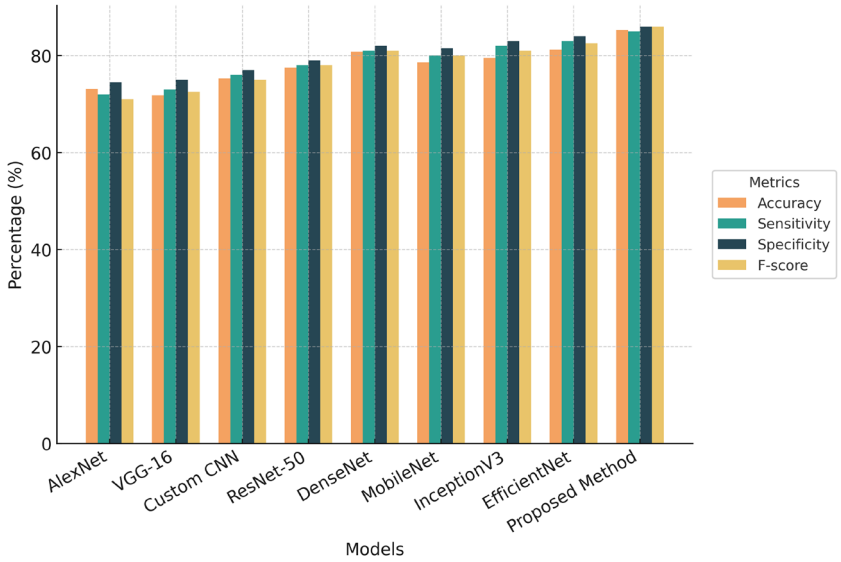

Fig. 9. Quantitative results on the New Mexico Decedent Image Dataset for the proposed model and deep learning models.

with computational efficiency, making them suitable candidates for real-world applications. The study also highlights the unique capability of Siamese NN to handle small datasets effectively without requiring retraining, making it highly advantageous for forensic scenarios.

Future research should aim to address the challenges posed by limited dataset sizes by expanding the dataset and incorporating advanced data augmentation techniques. Additionally, exploring ensemble methods that integrate the strengths of multiple models could further enhance the robustness and accuracy of forensic identification systems. This study establishes a foundation for leveraging deep learning to advance forensic practices, particularly in person identification from skull images.

Data availability

The raw code and dataset for our manuscript are available in the Github. The links are given below. - https://github.com/somer81/SiameseSkulls.

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Declarations

Competing interests

The authors declare no competing interests.

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